MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, Ⓔ RECEPTACLE, SOCKET CONTACTS, DECADE INCREMENTS 10 THRU 70 CONTACT POSITIONS AND INTERMEDIATE POSITIONS OF 14, 24, 44, 54 AND 26, 36, 56, 66 FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

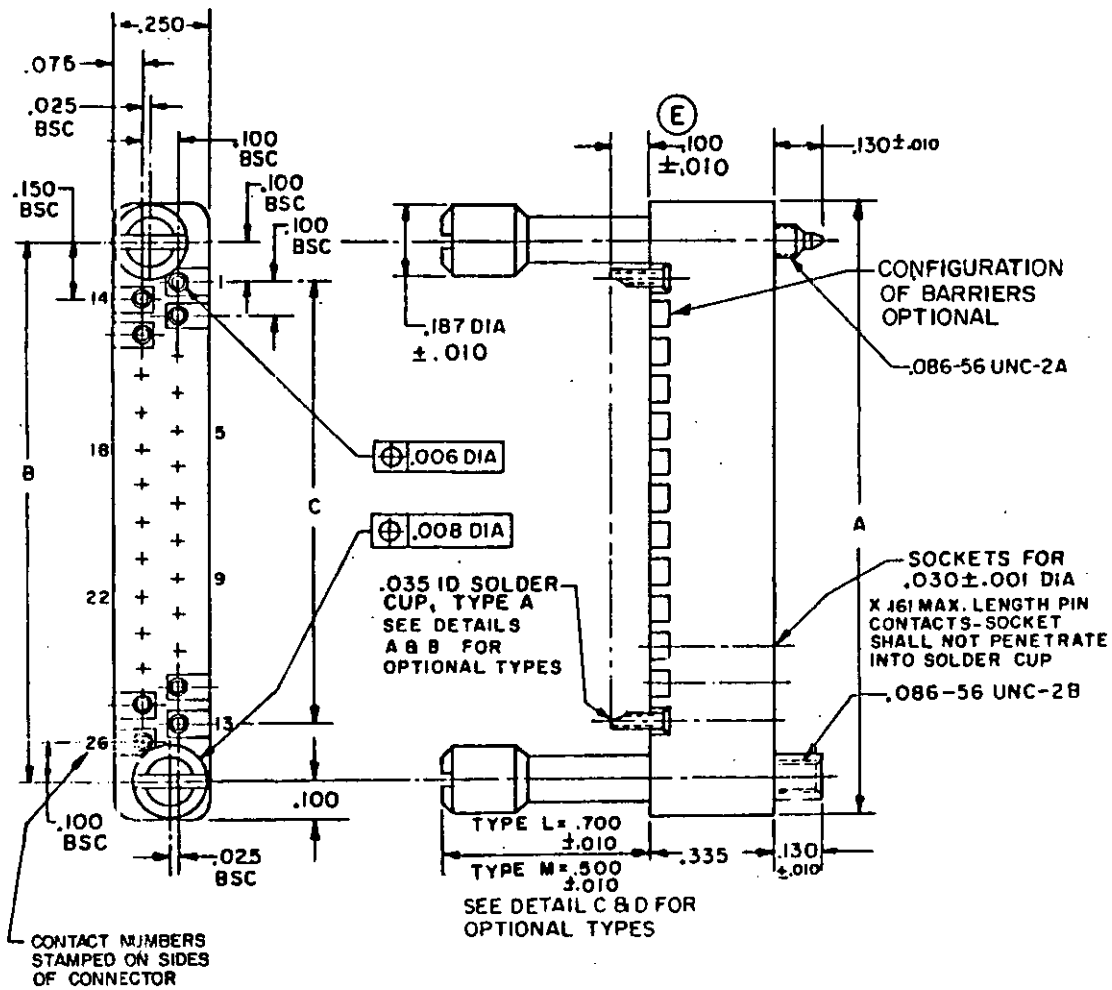
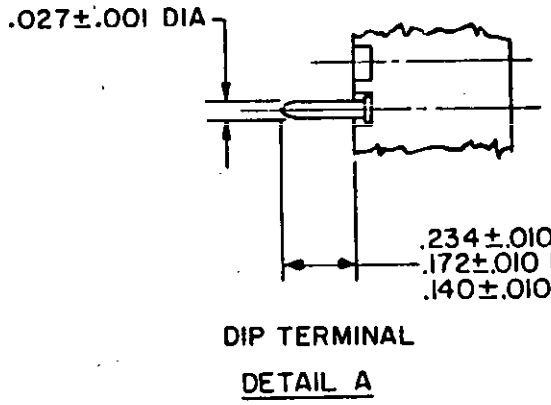


FIGURE 1. Connector, receptacle .100 (2.54 mm) spacing.

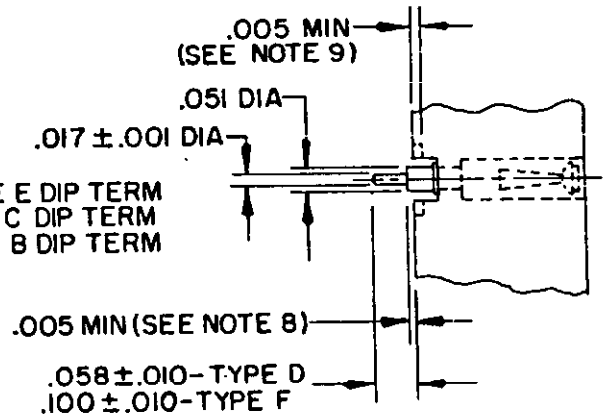
Ⓔ denotes changes

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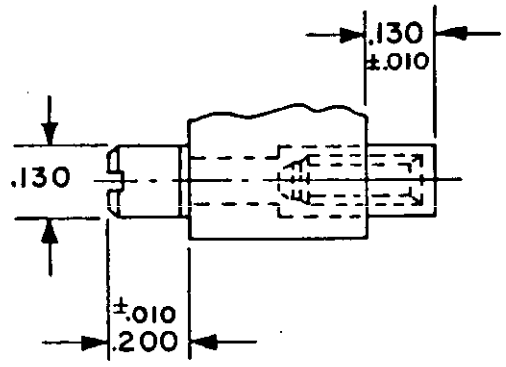
DIP TERMINAL

DETAIL A



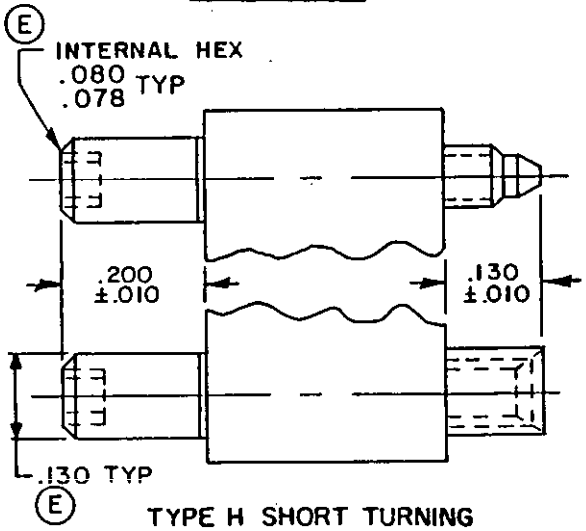
FLEXIBLE CIRCUIT TERMINAL

DETAIL B



TYPE S SHORT TURNING
JACKSCREW

DETAIL C



TYPE H SHORT TURNING
HEX JACKSCREW

DETAIL D

Number of contacts	Dimensions		
	A ±.010	BSC B	REF C
10	.850 (21.59)	.650 (16.51)	.400 (10.16)
14	1.050 (26.67)	.850 (21.59)	.600 (15.24)
20	1.350 (34.29)	1.150 (29.21)	.900 (22.86)
24	1.550 (39.37)	1.350 (34.29)	1.100 (27.94)
26	1.650 (41.91)	1.450 (36.83)	1.200 (30.48)
30	1.850 (46.99)	1.650 (41.91)	1.400 (35.56)
36	2.150 (54.61)	1.950 (49.53)	1.700 (43.18)
40	2.350 (59.69)	2.150 (54.61)	1.900 (48.26)
44	2.550 (64.77)	2.350 (59.69)	2.100 (53.34)
50	2.850 (72.39)	2.650 (67.31)	2.400 (60.96)
54	3.050 (77.47)	2.850 (72.39)	2.600 (66.04)
56	3.150 (80.01)	2.950 (74.93)	2.700 (68.58)
60	3.350 (85.09)	3.150 (80.01)	2.900 (73.66)
66	3.650 (92.71)	3.450 (87.63)	3.200 (81.28)
70	3.850 (97.79)	3.650 (92.71)	3.400 (86.36)

FIGURE 1. Connector, receptacle .100 (2.54 mm) spacing - Continued.

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Inches	mm	Inches	mm	Inches	mm
.001	0.03	.051	1.30	.171	4.09
.005	0.13	.058	1.47	.172	4.37
.006	0.15	.075	1.91	.187	4.75
.008	0.20	.078	1.98	.200	5.08
.010	0.25	.086	2.18	.234	5.94
.017	0.43	.100	2.54	.250	6.35
.025	0.64	.108	2.74	.335	8.51
.027	0.69	.130	3.30	.500	12.70
.030	0.76	.140	3.56	.700	17.78
.035	0.89	.150	3.81		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. These connectors mate with connectors specified in MIL-C-55302/56.
5. Numbers indicating the first and last position in each row and every fourth contact position in between shall be marked on the side(s) of the connector. As an option to the above, numbers indicating every fourth cavity may be stamped on the side of the connector, with the exception that the number one contact shall be marked. (E)
6. Termination layout on .025 (0.64 mm) modular grid.
7. Metric equivalents are in parentheses.
8. Conformal coating clearances shall be .005 (0.13 mm) minimum (not required for optional design).
9. Optional design conformal coating clearance.

FIGURE 1. Connector, receptacle .100 (25.4 mm) spacing - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Material: Jackscrews and jacksockets shall be made of corrosion resisting stainless steel per ASTM-A581 or ASTM-A582 passivated per QQ-P-35 or shall be corrosion resistant copper nickel alloy (61 ±2% copper, 25 ±2% nickel, 2.5% max. other, and the balance zinc) and shall have a brinell hardness of 145 to 175.

(E) Plating: The contact plating shall be per MIL-C-55302 or gold per MIL-G-45204, type II, grade C, class 1, over nickel plating per QQ-N-290, class 2, 50 to 150 microinches in the engaging area and gold per MIL-G-45204, type II, grade C, class 00, or type III, grade A, class 00, over nickel plating per QQ-N-290, class 02, 50 to 150 microinches in the termination area.

Contact identification: See figure 1.

(E) Contact engagement and separation forces (No. 22 contacts):
Minimum separation: 0.5 ounce.
Maximum engagement: 12.0 ounces for standard force and 4.0 ounces for low force.
Note: Low force contacts meet the requirements of standard force contacts.

Mating and unmating:
Standard insertion force contacts: The maximum mating force, in pounds, shall be the number of contacts multiplied by 0.56 and the withdrawal force, in pounds, shall be a minimum of 0.08 times the number of contacts and shall not exceed the measured insertion force.

Low insertion force contacts: The maximum mating force, in pounds, shall be the number of contacts multiplied by 0.25 and the withdrawal force, in pounds, shall be a minimum of 0.04 times the number of contacts and shall not exceed the maximum insertion force.

Contact resistance: The average contact resistance of all contacts measured shall not exceed 0.010 ohm, and no individual contact pair shall have a resistance exceeding 0.020 ohm.

Dielectric withstanding voltage:
Sea level: 1,000 volts rms.
High altitude: 300 volts rms.

Current rating: 5.0 amperes maximum per contact.

Example of part number:

M55302/55	L	A	10	L
Basic number of specification sheet	Type of contacts	Type of terminal (see figure 1)	Number of contacts (see figure 1)	Type of mounting hardware
	L = Low insertion force contacts - = Standard insertion force contacts	A = solder cup B = dip terminal .140 long C = dip terminal .172 long D = flexible circuit terminal.058 E = dip terminal .234 long F = flexible circuit terminal .100		L = .700 turning jackscrew slotted M = .500 turning jackscrew slotted S = .200 turning jackscrew slotted H = .200 turning jackscrew hex

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Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, TD
Air Force - 85, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC, OS
Air Force - 11, 19, 80

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3592-1)

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